Stats for Sociologists

Course: SOC6302H1F

Instructor: Professor Pepin

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Form Code: B

INSTRUCTIONS:

Fill out the first page of the scantron. Confirm you have included:

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1. A researcher compares support for universal healthcare (Support vs. Oppose) between two racial groups: Black and White respondents. Which statistical test should be used?
   (a) Chi-square test
   (b) t-test
   (c) Pearson's correlation (r)
   (d) Linear regression

2. Which command installs a new package in R?
   (a) `load()`
   (b) `require()`
   (c) `install.packages()`
   (d) `library()`

3. In order to compare meaningfully across groups, a frequency should be converted into a/an ____________.
   (a) standard deviation
   (b) percentage
   (c) number
   (d) ordinal variable

4. A sociologist wants to examine whether there is a relationship between level of education (measured in years) and annual income (in dollars) among adults in the U.S. Which statistical test is most appropriate?
   (a) Chi-square test
   (b) t-test
   (c) Pearson's correlation (r)
   (d) Logistic regression

5. Why is it considered good practice to use RStudio Projects when working on data analysis?
   (a) They help organize your files, set a working directory, and keep your environment consistent
   (b) They prevent syntax errors by enforcing tidyverse formatting rules
   (c) They ensure your code runs faster by optimizing memory usage
   (d) They automatically load packages when you open RStudio

6. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?
   (a) This couple had fewer arguments than about 2% of couples.
   (b) This couple had about 8 arguments per month.
   (c) This couple had fewer arguments than about 5% of couples.
   (d) This couple had fewer arguments than about 98% of couples.

7. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?
   (a) frequencies
   (b) cumulative frequencies
   (c) cumulative percentages
   (d) percentages

8. Which function is used to load a package in R?
   (a) `source()`
   (b) `install.packages()`
   (c) `library()`
   (d) `load()`

9. Which of the following is a valid use of `group_by()` and `summarize()`?
   (a) data |> summarize(mean = group_by(sex))
   (b) data |> group_by(sex) |> summarize(freq = n())
   (c) summarize(group_by(data, age), mean(sex))
   (d) group_by(data, summarize(sex))

10. Based on the table, which description best describes the correlation between personal income (`realrinc`) and number of days of poor mental health (`mntlhlth`)?

Correlation Coefficient	t-statistic	df	p-value	95% CI Lower	95% CI Upper
-0.11	-5	1933	0	-0.16	-0.07

   (a) Results indicate an inverse relationship (r = -0.11, p < .05), suggesting that the number of days of poor mental health decreases as personal income increases.
   (b) There is a weakly positive association between personal income and days of poor mental health (r = -0.11, p < .05).
   (c) Decreases in personal income was correlated with decreases in the number of days of poor mental health (t = -5, p < .05).
   (d) There is no evidence of an association between personal income and days of poor mental health (t = -5).

11. In *White Logic, White Methods*, what does Gandy suggest about the use of racial statistics?
   (a) They are neutral tools most useful for academic research
   (b) Their use in legal proceedings increasingly lead to unfair policy outcomes
   (c) They can both expose and perpetuate discrimination
   (d) Their use in data journalism helps mitigate online polarization

12. Using the table, which of the following is the correct formula to find the expected frequency for U.S. liberals who oppose the death penalty in 2024?

ideology	conservative	liberal	Total
cappun			
favor	512 (66.8%)	255 (33.2%)	767 (100.0%)
oppose	168 (32.7%)	345 (67.3%)	513 (100.0%)
Total	680 (53.1%)	600 (46.9%)	1280 (100.0%)

(a) (600 * 767)/1280
(b) (345 * 513)/1280
(c) (600 * 513)/1280
(d) (345 * 600)/1280

13. A researcher summarized the `sibs` variable from the U.S. General Social Survey, a measure of how many brothers and sisters respondents had. Based on the summary table, the distribution appears to be:

freq	median	mean	sd
3291	3	2.9	1.8

(a) negatively skewed
(b) positively skewed
(c) symmetrical
(d) nominal

14. According to Wheelan, in *Naked Statistics: Stripping the Dread from the Data?*, what is the most common cause of inaccurate polling results?

(a) Biased sampling methods or poorly worded questions
(b) Incorrect calculations of standard errors
(c) Random variation in respondent answers
(d) Misuse of statistical software during data analysis

15. What does the pipe operator (|>) do in dplyr?

(a) It filters rows based on a condition.
(b) It creates a new column in a data frame.
(c) It passes the result of one function into the next.
(d) It loads the dplyr package into memory.

16. In a normally distributed set of test scores, a student scoring below the average will have a ________________.

(a) positive z-score
(b) positive standard error
(c) negative standard error
(d) negative z-score

17. What will happen if you try to use a function from a package that hasn't been loaded?
    (a) R will automatically load the package
    (b) The function will run normally
    (c) R will prompt you to install the package
    (d) You will get an error saying the function is not found

18. What is the result of the following code?

```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```

    (a) Only rows with `age` > 20 and columns `age` and `childs` are returned.
    (b) rows with `age` > 20 are removed
    (c) The original gss24 dataset is overwritten.
    (d) A new column is added to `gss24`

19. Researchers calculated a 95% confidence interval of the average family size of Canadians to be from 2.7 to 3.1, based on a sample of 500 families with a standard deviation of 1.1. What change would produce a more precise estimate?
    (a) Increasing our confidence interval to 99%
    (b) Improving our sampling strategy
    (c) Increasing our standard deviation to 1.3
    (d) Increasing our sample size to 600

20. A respondent reports 5 poor mental health days in the past 30 days. Based on summary stats for the `mntlhlth` variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.5	7.9

    (a) The z-score is undefined because the median is 0.
    (b) The z-score is close to zero because 5 is close to the mean.
    (c) The z-score is negative because 5 is below the sd.
    (d) The z-score is large and positive because 5 is much higher than the mean.

21. What will the following code do?

```
gss24 <- gss24 |>
  mutate(status = case_when(
  income >= 100000 ~ "High",
  income > 50000 & income < 100000 ~ "Medium",
  TRUE ~ "Low"
))
```

(a) It removes missing values from the income column.
(b) It creates a new column named `status` with 3 possible values.
(c) It filters out rows where income is less than 50,000.
(d) It replaces the numeric values in the income column with text labels.

22. According to Cohen's argument in *Citizen Scholar*, what role should descriptive statistics play in social science research?

    (a) Descriptive statistics are useful only as a predominately as a step before hypothesis testing and theory development.
    (b) Descriptive statistics are a foundational part of scientific inquiry and should be valued as legitimate research.
    (c) Descriptive work should be minimized in favor of studies that make strong theoretical contributions.
    (d) Descriptive statistics are outdated and should be replaced by more advanced inferential methods.

23. A researcher suspects Americans are watching more tv today than in the 1970s. Assuming an α level of .05, should the researcher reject the null hypothesis? Why or why not?

t-statistic	df	p-value	95% CI Lower	95% CI Upper
-2.8	3617	0	-0.43	-0.08

    (a) Yes, because the p-value (0.005) is less than the α level of .05.
    (b) No, because the p-value (0.005) equals the α level of .05.
    (c) No, because the p-value (0.005) is greater than the α level of .05.
    (d) Yes, because the p-value (0.005) is greater than the α level of .05.

24. What is the purpose of `drop_na()` in dplyr?

    (a) It removes rows that contain missing values.
    (b) It converts NA values to NULL.
    (c) It drops columns that are entirely NA.
    (d) It replaces missing values with zeros.

25. The level of measurement for the `hapmar` variable is _________

    [1] "Happiness of marriage"

	Freq	%	% Cum.
very happy	802	59.36	59.36
pretty happy	504	37.31	96.67
not too happy	45	3.33	100.00
Total	1351	100.00	100.00

(a) nominal
(b) interval
(c) ordinal
(d) dichotomous

26. What does the following code return?

```
gss24 |>
  group_by(sex) |>
  summarize(mean_age = mean(age, na.rm = TRUE))
```

(a) A list of all ages grouped by sex.
(b) A new column named `mean_age` added to the original data frame.
(c) A filtered data frame with only rows where age is not missing.
(d) A data frame showing the average age for each sex.

27. In *Naked Statistics: Stripping the Dread from the Data?*, Wheelan explains that "most sample means will lie reasonably close to the population mean; the standard error is what defines 'reasonably close.' " What does this statement imply?

(a) The sample mean is always equal to the population mean.
(b) The standard error increases as sample size increases.
(c) The standard error tells us how far the population mean is from the sample mean.
(d) The standard error measures how much sample means tend to vary around the population mean.

28. Which of the following statements best reflects the ASA's guidance on interpreting p-values?

(a) A large p-value proves that no effect exists.
(b) p-values measure the scientific or economic significance of a result.
(c) A small p-value always indicates a large and important effect.
(d) p-values are influenced by sample size and measurement precision, and do not reflect effect size or importance.

29. Using 2024 GSS data, a researcher estimated the proportion of Americans who believe in life after death. Based on the 95% confidence interval from R, which interpretation is correct?

sex	n	mean	sd	lowCI	hiCI
male	1139	4.2	3.7	4.0	4.5
female	1174	4.0	3.5	3.8	4.2

(a) We cannot determine whether the null hypothesis of zero difference between the parameters will be rejected at the 0.05 level.
(b) The confidence intervals overlap, so we can't know for sure if there is a statistical difference between the average time men and women spend on leisure activities per day.
(c) We are 95% certain that women have less time per day than men for leisure activities, likely because of their greater time spent on housework and childcare.
(d) The confidence intervals do not overlap so we can conclude that the average time men and women spend on leisure activities per day are statistically significantly different.

30. Which of these datasets has a standard deviation of 0?

(a) 0, 2, 4, 6, 8
(b) 1, 4, 7, 10, 15
(c) 2, 2, 5, 7, 7
(d) 4, 4, 4, 4, 4

31. National survey data shows that 69% of Canadians support mandatory childhood vaccinations. A local researcher believes that support is likely much higher in Toronto compared with the national average. Which of the following equations represents the research hypothesis?

(a) $\mu = 0.69$
(b) $\mu < 0.69$
(c) $\mu \neq 0.69$
(d) $\mu > 0.69$

32. Which of the following code snippets correctly uses `mutate()` to convert height from centimeters to meters?

(a) mutate(height / 100 = height)
(b) mutate(height = select(height / 100))
(c) mutate(height <- height / 100)
(d) mutate(height = height / 100)

33. Which of the following is a measure of the spread of a distribution?

(a) standard deviation
(b) mean
(c) standardized variable
(d) median

34. Wheelan illustrates what key concept in the chapter "Inference: Why My Statistics Professor Thought I Might Have Cheated?" from the book *Naked Statistics: Stripping the Dread from the Data*?

    (a) That unusually accurate results can raise suspicion, but statistical inference helps assess whether they are likely due to chance
    (b) That hypothesis testing is not useful in criminal investigations
    (c) That cheating can be detected by comparing mean scores across different groups
    (d) That statistical inference can be used to prove someone's guilt beyond reasonable doubt

35. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?

    (a) Their score is unusual and higher than most.
    (b) Their score is very common.
    (c) Their score is within the middle 68% of the distribution.
    (d) Their score is likely below average.

36. Which description best describes a correlation coefficient of r = -0.62?

    (a) No relationship exists between the variables
    (b) Weakly negative relationship
    (c) A statistically insignificant relationship
    (d) Strong negative relationship

37. Using the U.S. General Social Survey data, a researcher tests whether the proportions who favor a law requiring a person to obtain a police permit before they could buy a gun have changed over time. Assuming an α level of .05, which of these statements best summarizes the researcher's conclusion?

t-statistic	df	p-value	95% CI Lower	95% CI Upper
1.8	3115	0.07	0	0.06

    (a) The is a statistically significant increase in support for the police permit law from 1984 to 2024, proving that stricter gun control laws are now favored by most Americans.
    (b) There is a statistically significant difference in support for the police permit law between 1984 and 2024, suggesting that public opinion has changed over time.
    (c) There is not a statistically significant difference in support for the police permit law between 1984 and 2024, indicating that public opinion has remained stable.
    (d) The difference in support for the police permit law between 1984 and 2024 was due to sampling error and cannot be interpreted.

38. Has public opinion shifted on whether Black Americans face worse jobs, income, and housing than white Americans due to discrimination? Using a significance level of $\alpha = 0.01$, what conclusion should be drawn from the test results?

	year		
	2022	2024	Total
opinion			
no	509 (45%)	522 (51%)	1,031 (48%)
yes	619 (55%)	494 (49%)	1,113 (52%)
Total	1,128 (100%)	1,016 (100%)	2,144 (100%)
Pearson's Chi-squared test, p=0.004			

(a) The p-value of .000 means the null hypothesis is true.
(b) The results show that most people in both years agreed racial inequality is due to discrimination.
(c) The evidence shows attitudes about racial inequality differ between 2022 and 2024.
(d) There is no statistically significant difference in attitudes between 2022 and 2024.

39. Which of the following best describes the differences between the Console, R Scripts, and Quarto documents in RStudio?

    (a) The Console is used for writing long-term code, R Scripts are for temporary testing, and Quarto is only for plotting.
    (b) The Console and R Scripts are identical, while Quarto is used only for Markdown formatting.
    (c) R Scripts and Quarto documents both run code line-by-line, but only the Console saves your work.
    (d) The Console runs code interactively, R Scripts store reusable code, and Quarto combines code, text, and output for dynamic reports.

40. If a researcher concluded that there was no difference in police use of force against Black and White men when there actually is a difference, then:

    (a) The null hypothesis was correctly rejected.
    (b) The null hypothesis was correctly retained.
    (c) The null hypothesis was incorrectly rejected (false positive).
    (d) The null hypothesis was incorrectly retained (false negative).

41. If the final grades in a statistics class were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?

    (a) 62%
    (b) 50%
    (c) 68%
    (d) 95%

42. Chi-square values cannot be _____.
    (a) negative
    (b) positive
    (c) zero
    (d) outside the range of 0 to 1

43. What does the expression `5 != 3` evaluate to in R?
    (a) FALSE
    (b) NULL
    (c) NA
    (d) TRUE

44. What does the `::` operator do in R?
    (a) Comments out a line of code
    (b) Imports all functions from a package
    (c) Assigns a value to a variable
    (d) Specifies a function from a particular package

45. What does the following R code produce?

```
gss21 |>
  tbl_cross(
    row = incgap,
    col = sex,
    percent = "column",
    missing = "no") |>
  add_p()
```

    (a) A cross-tabulation of `incgap` by `sex`, showing column percentages and a chi-square test of independence
    (b) A regression output predicting `incgap` from `sex`, including R-squared and p-values
    (c) A summary table comparing the mean values of `incgap` between men and women using a t-test
    (d) A frequency table of `incgap` responses with missing values imputed and grouped by `sex`

46. Based on the table, which statement best compares the relative frequency of being 'not too happy' between men and women?

sex	male	female	Total
happy			
very happy	293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy	830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy	330 (22.7%)	370 (20.4%)	700 (21.5%)
Total	1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

(a) The frequency of being 'not too happy' is equal for men and women.
(b) Women have a slightly higher relative frequency because more women than men reported being 'not too happy'.
(c) A greater number of women reported being 'not too happy' than men, so women have a higher relative frequency.
(d) Men have a slightly higher relative frequency of being 'not too happy' than women, even though fewer men reported it.

47. Which RStudio pane is used to write and edit R scripts and Quarto documents?
    (a) Source
    (b) Environment
    (c) Console
    (d) Files

48. Using 2024 GSS data, a researcher estimated how many Americans believe in life after death. Based on the 90% confidence interval from R, which interpretation is correct?

n	mean	sd	lowCI	hiCI
1283	0.83	0.38	0.81	0.85

    (a) The researcher is 90% confident that the proportion of U.S. adults who believe in some form of life after death is between .81 and .85.
    (b) If another survey had been conducted in 2024, there's a 90% chance the point estimate would be between .81 and .85.
    (c) About 90% of people responded to the national survey.
    (d) Because the mean is within the confidence interval, there is a .90 probability that the proportion of U.S. adults who believe in some form of life after death is .83.

49. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data?*
    (a) Statistics provide honest answers to complex questions
    (b) Statistics explain why relationship exist between variables
    (c) Statistical formulas are the most important part of understanding data
    (d) Statistics simplify complex information to help us make better decisions

50. A p-value is
    (a) A measure that tells us the variability in a normal distribution
    (b) An estimate of where our population statistic lies on the true sampling distribution of the means
    (c) An estimate of the standard error of a sampling distribution
    (d) A measure of how unusual or rare our obtained statistic is compared with what is stated in the null hypothesis

Answer Sheet

	(a)	(b)	(c)	(d)
1.	X			
2.			X	
3.		X		
4.			X	
5.	X			
6.				X
7.			X	
8.			X	
9.		X		
10.	X			
11.			X	
12.			X	
13.	X			
14.	X			
15.			X	
16.				X
17.				X
18.	X			
19.				X
20.		X		

	(a)	(b)	(c)	(d)
21.		X		
22.		X		
23.	X			
24.	X			
25.			X	
26.				X
27.				X
28.				X
29.		X		
30.				X
31.				X
32.				X
33.	X			
34.	X			
35.	X			
36.				X
37.			X	
38.			X	
39.				X
40.				X
41.			X	

42.	(a)	**X**	(b)		(c)		(d)	
43.	(a)		(b)		(c)		(d)	**X**
44.	(a)		(b)		(c)		(d)	**X**
45.	(a)	**X**	(b)		(c)		(d)	
46.	(a)		(b)		(c)		(d)	**X**
47.	(a)	**X**	(b)		(c)		(d)	
48.	(a)	**X**	(b)		(c)		(d)	
49.	(a)		(b)		(c)		(d)	**X**
50.	(a)		(b)		(c)		(d)	**X**